PathVeil: Learning AS Routes from Public Evidence without Source Measurements

Anonymous Authors

***Abstract*—Learning how an autonomous system (AS) reaches remote networks usually requires observations of routes originating from that AS. Such observations are unevenly available, leaving a source cold start for path inference. We present PathVeil, a framework that constructs source-specific training paths from other networks' routing observations and public interconnection evidence. Its prefix-anchored tail stitching method selects likely forwarding neighbors, resolves routing choices beyond the first hop when necessary, and joins selected prefixes to observed route segments. A learned complete-path scorer admits a source-specific library using relative quality, while the same selected prefix tree constrains training and query candidate generation. A GMPI model learns from this library to rank candidate routes. On three monthly BGP snapshots, each containing ten target sources and more than 685,000 query pairs, PathVeil obtains 82.56%, 85.06%, and 83.45% AS-set Jaccard. Source-level diagnostics distinguish first-hop coverage from complete-path observation support and expose cases where correct first-hop sets still provide weak training paths. These results assess route agreement under the stated limits of public evidence.**

***Index Terms*—AS path inference, source cold start, path augmentation, interdomain routing, public routing evidence**

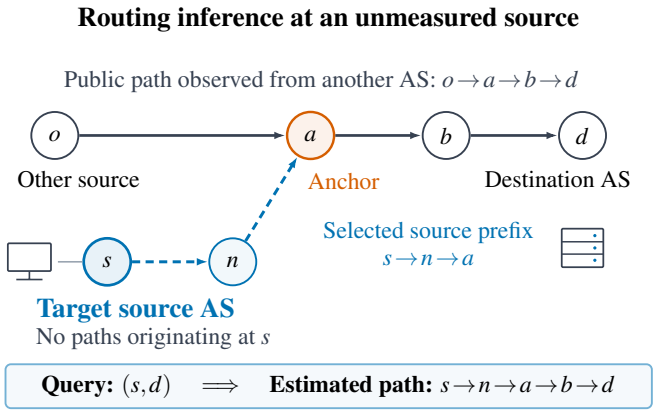


Fig. 1. A selected source prefix can connect an unmeasured source to a publicly observed tail. Dashed arrows denote the inferred prefix; solid arrows denote observed segments. Construction supplies a candidate, not proof that the source forwards along it.

I. Introduction

Internet services depend on routes through autonomous systems (ASes) that users and service operators do not directly control. Knowing which networks a route traverses can help identify transit dependencies, inspect exposure to intermediary networks, and compare connectivity to candidate service locations [1]. These tasks require path visibility from the network of interest, rather than only from a nearby measurement probe. For example, a source AS s may reach many destinations through one neighbor, while the important routing choices occur at that neighbor. Figure 1 illustrates the problem: public observations contain useful routes beyond s, but do not directly disclose which continuations s would use.

Direct path measurements provide observations from deployed vantage points. Traceroute requires a probe in, or suitable access to, the source network; distributed platforms improve coverage but cannot guarantee that such a probe exists for every AS [2], [3]. Public BGP collectors supply routes from participating networks, whose observations are also uneven [4], [5]. Path inference extends this coverage, but the information it needs depends on the method. Graph-based approaches can search for routes using public topology and policy approximations, although these rules cannot capture all local preferences [6], [7]. Stitching-based approaches reuse observed segments but require evidence that connects the source to those segments [8]. Learning-based approaches such as GMPI benefit from many source-rooted paths to learn source-specific routing behavior [9], [10].

When observations originating from the target AS are unavailable, these approaches face a source cold start. We address it by constructing source-rooted training paths from routing evidence collected elsewhere. Observed tails preserve routing decisions already made by other networks; public interconnection information helps infer the short prefix connecting the source to those tails. The resulting library can support learning and candidate generation across destination queries. The aim is to infer routes without collecting new measurements from the target source, while making explicit which parts of a route remain inferred.

Three challenges govern this construction. **First, where should source-specific exploration occur?** A well-observed neighbor supplies reusable tails directly; a poorly observed neighbor may require exploration beyond that neighbor. A source with few neighbors can also hide substantial routing diversity at its second hop. Selecting likely forwarding neighbors and deciding whether to expand them therefore form one problem. **Second, how should paths be constructed and admitted?** A graph route can satisfy coarse policy rules without reflecting an observed routing decision. Stitching retains observed decisions, but an otherwise plausible join can still produce a route that the source would not use. Candidate construction must be coupled with complete-path assessment, and the admitted library must be examined for concentration on a few easy branches. **Third, how can heterogeneous**

public evidence support these choices? Relationships, topology, BGP segments, and co-location metadata have different meanings and coverage. They must inform AS, prefix, and path representations without treating missing information as a negative observation or shared facilities as established links.

We present **PathVeil**, whose core method is **prefix-anchored tail stitching (PATS)**. **For the first challenge,** source-topology-conditioned prefix selection combines learned neighbor ranking with bounded exploration. A source-shape statistic compares distinct two-hop reachability with first-hop degree. For one regime, usable-tail capacity determines whether to explore further; for the other, every selected first hop is expanded to resolve downstream choices. The output retains complete prefixes and their commercial-phase context, forming one source-specific tree shared by subsequent stages.

For the second challenge, observed-tail construction and learned quality admission connect short selected prefixes to intact forward segments. This reuses much of an existing routing decision instead of inventing the full route. A complete-path model assesses support using prefix, segment, and destination context. Admission uses a source-relative score threshold with an explicit minimum-library rule. During inference, the selected tree anchors both observed tails and topology-derived alternatives. This design controls where routes can leave the source; it does not assume that a quality threshold alone guarantees branch representation.

For the third challenge, a multilevel evidence representation encodes network attributes and co-location, directed transitions and commercial phases, and complete-path support. Destination-conditioned features compare observations across independent donor sources, with per-donor normalization to reduce domination by prolific observers. Unknown relationships and missing evidence remain explicit. The admitted path set supplies both source-specific training labels and local path knowledge for candidate generation, keeping these consumers consistent. We use GMPI as the downstream ranking model without changing its architecture or learning objective.

Our contributions are:

- A framework for source-cold-start AS route inference that builds source-specific learning evidence from public observations and interconnection metadata, without requiring measurements originating from the target source as an input.
- A prefix-anchored path-augmentation method that combines source-topology-conditioned exploration, reuse of observed tails, and complete-path quality admission, with one prefix tree shared by training and query generation.
- An evaluation of constructed-source inference on fixed BGP query cohorts, with source-level results and distinct criteria for prefix selection, augmented-path observation support, and final route agreement. The evaluation distinguishes representation through selected neighbors from complete-path observation support and final AS-set agreement.

II. Background and Related Work

A. Routing Observations and Their Limits

BGP separates learning a route from a neighbor, selecting a route under local policy, and exporting it to other neighbors [11]. Consequently, an observation of $(n, \ldots, d)$ does not establish that n exports that route to source s, or that s selects it. Route Views and RIPE RIS expose advertised AS paths from participating networks [4], [5]. CAIDA relationships and ASRank add commercial roles, connectivity, and customer-cone attributes; PeeringDB supplies self-reported network and interconnection metadata [12]–[14]. These sources constrain plausible routing but do not reveal private preferences. Shared IXP or facility presence does not prove adjacency. Valley-free compatibility describes a common commercial model, not a complete account of operational policy [15].

Traceroute discovers responding hops by varying IP time-to-live. Classic traceroute can combine replies from multiple paths under load balancing; Paris traceroute controls flow identifiers to reduce per-flow artifacts [2], [16]. Scamper supports parallel, rate-controlled measurement, and RIPE Atlas provides distributed probes [3], [17]. Their output still depends on source placement, filtering, and responding routers. Unresponsive hops leave gaps, and IP-to-AS conversion introduces additional ambiguity [18], [19]. A traceroute data-plane observation and a BGP control-plane path are related evidence, rather than interchangeable ground truth.

B. AS Path Inference

KnownPath uses BGP paths as anchors for inferring source-to-prefix routes under routing constraints [6]. **ProbInfer** represents observed path segments in a multigraph, composes candidates, and evaluates them probabilistically [8]. These approaches demonstrate the value of preserving observed segments; their source connectivity still depends on the available anchors and policy assumptions.

GMPI combines heuristic path generation with a neural evaluation model that integrates AS attributes, topology, and related observed paths through path representations and attention [9]. **PA-GMPI** adds personalized generation and adaptive training, including a graph-kernel warm starter and representations learned from observed walks [10]. Both motivate source-specific learning, while their training observations are an important resource requirement. PathVeil supplies constructed source-specific evidence and a common prefix constraint; GMPI provides the downstream candidate score.

PathRadar distinguishes core and shell ASes using routing complexity and observation adequacy, combining policy-based and stitching-based inference. Progressive learning propagates candidate routes and addresses prefix-specific and multiple-path inference [7]. Its treatment of heterogeneous observation conditions is related to our exploration decision. Our method uses a selected source prefix tree and complete-path admission to build a reusable training library.

Other systems address adjacent objectives. iPlane composes measured segments to predict paths and their properties [20],

[21]; PredictRoute combines historical control- and data-plane evidence for probabilistic prediction [22]; RouteInfer models commonality and diversity in ISP routing [23]. A broader survey reviews the applications and limitations of interdomain path inference [1]. These works provide context for our focus: learning routes for a target source whose own originating observations are unavailable to construction.

III. Problem Setting and Evidence Roles

A. Source-Cold-Start Queries

A query specifies a source AS s, destination AS d, and snapshot t. We return an ordered forward AS sequence $\widehat{P}_{s,d}$, or no usable estimate. The source need not host a probe or contribute source-originated paths to the construction input. Observations from other origins may contain s internally; they remain indirect public evidence. We never reverse an observation to manufacture a forward route.

The AS-pair abstraction pools destination prefixes within an AS. Different prefixes can follow different routes, so one returned sequence is an estimate rather than a guarantee for every packet or endpoint. For a service query, resolving an endpoint and mapping its prefix to d precedes our inference; DNS steering and anycast can change that endpoint or its serving instance [24], [25]. We do not infer the return path, router sequence, geography, or latency.

B. Public Evidence and Supervision

We distinguish four roles. Public background paths supply topology and reusable observed segments. Reference paths from separate training origins supervise the neighbor and path-support models. Augmented paths form the target source's downstream training library. Finally, held-out source routes label final evaluation queries and do not supervise target construction or checkpoint selection.

Let $\mathcal{G}_t$ be the public AS graph and $\mathcal{P}_{\mathrm{pub}}$ the permitted background path corpus. A path $P = (v_0, \ldots, v_\ell)$ supplies a forward tail $(v_i, \ldots, v_j)$ for $i < j \leq \ell$. A selected prefix $q = (s, \ldots, v_i)$ joins that tail at v_i, including the anchor once. Reuse preserves the observed order. An internal endpoint $j < \ell$ is allowed; it supplies evidence to the AS at that endpoint. Training-donor observations additionally support destination-context features and inference candidates under their own origin restrictions. These roles are not merged simply because two files contain the same AS. Table I summarizes the external data.

C. Library and Inference Objective

Construction produces a selected prefix tree $\mathcal{T}_s$ and an admitted library $\mathcal{L}_s$. The tree retains the source, each chosen first hop, any explored second hop, and feasible commercial-phase states. Both training and query candidate generation use this complete tree. The same ordered path set $\mathcal{L}_s$ provides downstream training labels and local path knowledge, verified by a common path-set hash.

For candidate set $\mathcal{C}_{s,d}$ and available history $H_{s,d}$, query inference selects

$$\widehat{P}_{s,d} = \operatorname*{arg\,max}_{P \in \mathcal{C}_{s,d}} f_{\theta_s}(P, H_{s,d}). \tag{1}$$

The source model f_{θ_s} is trained on $\mathcal{L}_s$. The complete-path admission score and the query score in (1) serve different decisions; neither is interpreted as a calibrated probability of actual forwarding.

IV. Methodology

A. Framework Overview

PathVeil addresses source cold start by transferring routing evidence from other networks. Its augmentation procedure, PATS (Prefix-Anchored Tail Stitching), first selects likely forwarding neighbors and determines whether to explore their next hops. It then joins the resulting prefixes to intact observed tails and assesses the complete constructed routes. The resulting library trains a source-specific inference model without opening the source's held-out route labels. Figure 2 shows the separation between path augmentation and final inference.

Two objects are retained for each source: the complete selected prefix tree $\mathcal{T}_s$ and the admitted path library $\mathcal{L}_s$. They serve different purposes. The tree defines the permitted early forwarding choices; the library provides surrogate training routes and local path knowledge. Candidate generation uses the complete tree, including selected prefixes that do not happen to dominate the admitted library. Thus library selection cannot silently redefine the source's permitted exits.

B. Representing Heterogeneous Routing Evidence

We represent public AS relationships and ASRank attributes, PeeringDB presence, and independent BGP observations at AS-pair, transition, and complete-path levels [13], [14], [28]. Table II summarizes their roles. Features are tabular, with explicit missingness and conflict indicators. Shared PeeringDB IXPs or facilities identify potential interconnection partners; co-location alone does not establish a physical link or an export policy.

a) Separating evidence from supervision.: The upstream procedure masks complete observations by their original first ASN before expanding paths into segments. Independent background observations provide ordered edges and forward contiguous segments. Task-source observations have separate roles: training origins provide supervised labels and destination-conditioned donor statistics, while held-out origins provide evaluation labels. A segment containing s inside another origin's observation can remain evidence; an observation originally rooted at held-out s is excluded from this upstream evidence. Tails preserve their observed direction and are never constructed by reversing a route. These upstream guarantees apply to the explicitly masked corpus; the provenance of inherited downstream graph and embedding artifacts is assessed separately in the experimental protocol.

TABLE I
EXTERNAL DATA AND THEIR ROLES. MONTHLY SNAPSHOTS REFER TO FEBRUARY 1, MARCH 1, AND APRIL 1, 2026. EVIDENCE AND FINAL REFERENCE LABELS HAVE SEPARATE CONSUMERS.

Source	Snapshot / period	Contents and use
Route Views / RIPE RIS [4], [5]	Three monthly dates; Oct. 15	Public AS paths for topology, observed segments, and separately partitioned upstream supervision. Designated source routes label the fixed BGP query cohorts. Inherited graph-feature provenance is discussed in the evaluation protocol.
CAIDA relationships [12]	Three monthly dates; Oct. 1	Directed provider–customer and peer roles for AS/prefix/path features and commercial-phase checks. Missing roles remain unknown.
CAIDA ASRank [13]	Three monthly dates; Oct. 1	AS and link attributes, including rank, degree, and customer-cone size, for neighbor ranking and public graph representation.
PeeringDB [14]	Three monthly dates; Oct. 14	Network attributes and IXP/facility memberships for candidate retrieval and co-location features; shared presence does not establish adjacency.
Tranco N26XW [26]	Oct. 14	Frozen one-million-domain list for the independent endpoint dataset.
DNS resolution (this study)	Oct. 15	911,257 resolved domains; 388,491 unique IPv4 probe targets.
CAIDA prefix-to-AS [27]	Oct. 13, 10:00 UTC	Longest-prefix origin mapping for endpoints and responding hops; ambiguous mappings do not become certain labels.
Traceroute (this study)	Oct. 15–16	AS63835: 388,491 TCP and 243,207 ICMP Paris records. Filtering yields 2,198 distinct ordered AS paths over 1,890 pairs for evaluation only.

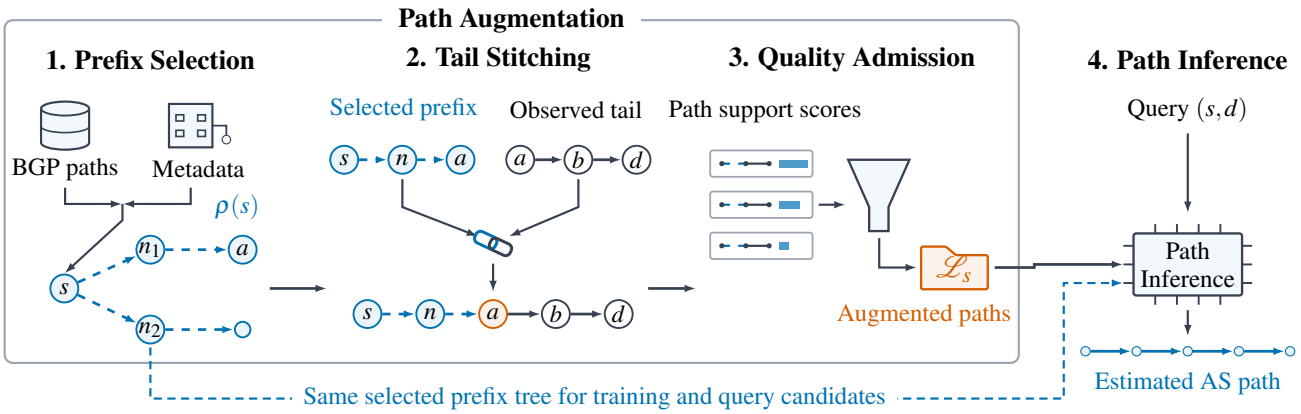


Fig. 2. PathVeil constructs a source-specific prefix tree, joins its prefixes to observed tails, and selects an augmented path library. The same complete tree constrains candidate generation during model training and inference; the admitted library supplies both training labels and local path knowledge.

b) Destination-conditioned transfer.: A path observed at an intermediate AS need not be exported to a particular neighbor [11]. We therefore complement segment support with evidence about which next hops other receivers observe toward the same destination. For an original training donor v, let $c_v(n,a,d)$ count distinct complete routes beginning (v,n,a) and ending at d, and let $c_v(n,d) = \sum_b c_v(n,b,d)$. With $V_{n,d} = \{v : c_v(n,d) > 0\}$, two representative features are

$$f(n,a,d) = \frac{|\{v \in V_{n,d} : c_v(n,a,d) > 0\}|}{|V_{n,d}|},$$
$$m(n,a,d) = \frac{1}{|V_{n,d}|} \sum_{v \in V_{n,d}} \frac{c_v(n,a,d)}{c_v(n,d)}. \quad (2)$$

The first measures agreement across donors and the second gives every donor equal weight after normalizing its alternative routes. We also condition these statistics on the donor's incoming commercial role at n, matching the candidate receiver's role when it is known. This uses original receiver contexts, rather than treating every internal occurrence of n as an independent receiver observation.

These features are evaluated at both the first-hop transition and the construction boundary. Two 16-column context vectors and one boundary indicator add 33 columns to 181 path-evidence, 32 prefix-reliability, and nine actual-next-hop columns, giving a 255-column complete-path representation. Missing donor coverage remains explicit. For a training source, the entire original-source fold is removed from its donor statistics, even when recursion changes the stitching anchor.

C. Source-Conditioned Neighbor Selection and Exploration

a) Learning likely forwarding choices.: Candidate neighbors are retrieved from public or independently observed adjacency and PeeringDB co-location, with graph-supported candidates prioritized in a pool of at most 4,096 ASes. Two LightGBM rankers [29] use a public-evidence view and a structurally extended view. The latter adds common-neighbor overlap, Adamic–Adar, resource-allocation, and population-

TABLE II
EVIDENCE REPRESENTATIONS AND THEIR COMPUTATIONAL ROLES. MISSING AND CONFLICTING OBSERVATIONS REMAIN EXPLICIT.

Representation	Features and use
AS pair	Public adjacency and commercial roles; ASRank degree, cone, and network attributes; PeeringDB IXP/facility overlap and population; independent edge support. Neighbor retrieval and ranking.
Graph structure	Common-neighbor overlap, Adamic–Adar, resource allocation, role-conditioned bridges; source first- and second-layer size. Structural ranking and $\rho(s)$.
Prefix/transition	Preceding, current, and next AS; local rank and presence scores; edge provenance and possible commercial phases. Conditional exploration and prefix-state tracking.
Path evidence (181)	Ordered path, suffix, edge and destination support; AS count and commercial-role sequence; public endpoint context. Base features for complete-path scoring.
Prefix reliability (32)	Support/missingness across added edges, inferred-edge fraction, feasible phase states, and soft first-hop transit-role evidence.
Actual next hop (9)	Conditional support, rank, missingness, and phase status of the candidate's actual third AS.
Destination context (33)	Donor-source agreement and equally weighted route fractions at the first hop and stitching boundary, with incoming-role conditioning and explicit absent-boundary indicator.

weighted IXP/facility features. Each view also has a binary presence head.

Training first-hop importance gives each destination equal weight, divided among its distinct observed routes. Observed neighbors ranked 1–3, 4–10, and beyond ten receive relevance labels 3, 2, and 1; unobserved candidates receive 0. Presence indicates use in this finite reference corpus. These targets measure observed routing choices, rather than link existence or traffic volume.

Let $r_s^{\mathrm{pub}}(n), r_s^{\mathrm{str}}(n)$ denote the ranker outputs and $p_s^{\mathrm{pub}}(n), p_s^{\mathrm{str}}(n)$ the presence outputs. The protected core is

$$K_s = \mathrm{Top}_3(r_s^{\mathrm{pub}}) \cup \mathrm{Top}_3(r_s^{\mathrm{str}}). \tag{3}$$

For ordering within the core and the remaining candidates, we use

$$\begin{aligned} \bar{p}_s(n) &= \tfrac{1}{2}\big(p_s^{\mathrm{pub}}(n) + p_s^{\mathrm{str}}(n)\big), \\ u_s(n) &= z(r_s^{\mathrm{str}}(n)) + 0.5 \log \frac{\bar{p}_s(n)}{1-\bar{p}_s(n)}, \end{aligned} \tag{4}$$

where z standardizes within the candidate pool and presence outputs are clipped to $[10^{-6}, 1-10^{-6}]$ before taking the logit. The core precedes the remaining candidates, with descending u_s and ASN tie breaking. Presence outputs are uncalibrated observation-support scores; we do not interpret them as forwarding probabilities.

b) Adapting exploration to source structure.: Degree alone misses a source whose first hop conceals a large next-layer choice set. Let $b_1(s)$ be the source's public degree, and let $b_2(s)$ count distinct second-layer ASes reachable through loop-free, commercially non-incompatible two-hop public prefixes. Unknown roles remain eligible. We define

$$\rho(s) = \frac{b_2(s)}{b_1(s)^2}. \tag{5}$$

This is a structural control statistic, not a routing probability. It is computed once for the original source and is not recomputed to switch policies at its neighbors. A source with no usable public degree remains unresolved.

When $\rho(s) \le 1$, we select the longest prefix of the protected-core ordering containing K_s, limited to a root budget κ, whose mean $\bar{p}_s$ is at least 0.7. If none qualifies, the core is retained with uncertain support. For each selected neighbor n, we count usable independent forward tails for (s, n), stopping once 300 are found. A usable direct prefix is retained. If its capacity is below 300, exploration also proposes (s, n, a) prefixes. An explored child is retained only if it has at least one usable tail. The evaluated protocol uses $\kappa = 21$, the 90th-percentile number of first-hop neighbors needed to cover 80% of original training paths in the February source statistics. This is a search budget, not a guarantee of reference coverage for a new source.

When $\rho(s) > 1$, only the dual-ranker core K_s is selected, yielding at most six roots. Every root is expanded to (s, n, a) independently of its direct-tail capacity. This exposes choices hidden behind a narrow first layer; the high-ρ branch does not retain a broad direct-tail bypass. Both policies use the same next-hop width of ten and a maximum prefix length of three ASes.

c) Prefix-aware next-hop prediction.: A conditional ranker uses the preceding AS, current AS, and candidate next AS to transfer local routing preferences. Its candidate pool is limited to 512. Each proposed extension must avoid visited ASes and have independent/public edge evidence, or co-location accompanied by a finite learned support score. It must also preserve at least one feasible commercial phase of the complete prefix. We track upward, post-peer, and downward states; sibling edges are neutral and unknown relationships retain all feasible states. Distinct prefixes and phase states sharing an anchor remain distinct. This is local conditional transfer, not a claim that multiplying local scores gives a calibrated path probability. No supported continuation means stopping that branch rather than inventing an extension.

D. Prefix-Anchored Tail Construction

For each selected prefix $q = (s, \ldots, z)$, PATS joins it to an intact observed forward tail $t = (z, \ldots, d)$:

$$q \oplus t = (s, \ldots, z, \ldots, d), \tag{6}$$

including the common anchor z once. Reusing a tail preserves multiple consecutive observed routing choices. It does not establish that s would learn or select the resulting route.

Construction checks valid ASNs, 3–20 ASes, absence of loops, support for each added prefix edge, and existence of a valley-free assignment consistent with known relationships [15]. Missing relationships are retained as unknown

rather than automatically rejected. The same physical path may arise from different prefixes; ordered sequences are deduplicated while retaining their construction provenance.

To bound scoring work while preserving alternatives, construction keeps up to two evidence-prioritized candidates per first hop, actual third AS, destination, and AS count. It favors fewer unobserved prefix extensions and stronger independent segment support. A deterministic hash ordering selects 5,000 destination contexts, supplemented by all destinations of direct prefixes covering at most 1,000 destinations and up to 50 protected destinations per other explicit prefix. These are construction work budgets, not a final library-size cap.

E. Complete-Path Scoring and Library Admission

a) Learning finite-observation support.: A LightGBM classifier $h_\theta(\phi(P))$ assesses the 255-column representation of a complete constructed path. For a training source, an exact ordered match to its original reference is positive. A nonmatching candidate is negative only if its destination has reference coverage; candidates for uncovered destinations remain unlabeled. The weighted binary objective balances training sources, first hops, destinations within first hops, and alternatives within each context. The model uses 300 trees, at most 31 leaves per tree, maximum depth seven, and learning rate 0.04.

Models are fitted separately for each date using registered training origins. Neighbor and transition predictions for a training source use heads fitted without its original-source fold; destination features exclude the same fold. This is stagewise source isolation. The final path classifier is fitted on these training cases, and held-out source routes never determine its features or parameters. Because non-observation does not prove route impossibility, h_θ estimates support in a finite observation corpus, rather than the probability of forwarding reality.

b) Source-relative admission.: Let $\mathcal{B}_s$ be the legal deduplicated source candidate pool and $q(P) = h_\theta(\phi(P))$. We normalize within that source:

$$\widetilde{q}_s(P) = \frac{q(P)}{\max_{P' \in \mathcal{B}_s} q(P')}. \tag{7}$$

The nominal threshold is 0.75. To retain at least 1,000 paths when positive-scored capacity permits, let $\widetilde{q}_{s,(1000)}$ be the 1,000th highest relative score and set

$$\begin{aligned} \tau_s &= \min\{0.75, \widetilde{q}_{s,(1000)}\}, \\ \mathcal{L}_s &= \{P \in \mathcal{B}_s : \widetilde{q}_s(P) \geq \tau_s\}. \end{aligned} \tag{8}$$

All cutoff ties are retained and there is no upper size cap. The library is unresolved if fewer than 1,000 legal paths exist, if the required cutoff is zero, or if the source maximum is zero. The threshold relaxation and actual cutoff are recorded. Relative quality is an admission ranking criterion, not a guarantee of 75% route accuracy. Algorithm 1 summarizes augmentation.

Algorithm 1 Prefix-Anchored Tail Stitching and Admission

Require: Source s, public topology G, independent tails $\mathcal{T}$
Require: Neighbor heads g, conditional ranker g_c, path scorer h_θ
Require: Root budget κ (21 in the evaluated protocol)
Ensure: Prefix tree $\mathcal{T}_s$ and admitted library $\mathcal{L}_s$, or unresolved

if $b_1(s) = 0$ **then**
  return unresolved
end if
$\rho \leftarrow b_2(s)/b_1(s)^2$
$K \leftarrow \mathrm{Top}_3(g_{\mathrm{pub}}(s)) \cup \mathrm{Top}_3(g_{\mathrm{str}}(s))$
if $\rho \leq 1$ **then**
  $N \leftarrow \textsc{ProtectedSupportPrefix}(s, K, \kappa, 0.7)$
else
  $N \leftarrow K$
end if
$Q \leftarrow \emptyset$
for all $n \in N$ **do**
  $q \leftarrow (s, n)$; $c \leftarrow \textsc{UsableTailCount}(q, \mathcal{T}, 300)$
  if $\rho \leq 1$ and $c > 0$ **then**
    $Q \leftarrow Q \cup \{q\}$
  end if
  if $\rho > 1$ or $c < 300$ **then**
    $A \leftarrow \textsc{SupportedTopK}(g_c(s, q), 10)$
    for all $a \in A$ with $a \notin q$ **do**
      $q' \leftarrow q \parallel (a)$
      if $\rho > 1$ or $\textsc{UsableTailCount}(q', \mathcal{T}, 1) > 0$ **then**
        $Q \leftarrow Q \cup \{q'\}$
      end if
    end for
  end if
end for
$\mathcal{T}_s \leftarrow \textsc{PreservePrefixPhaseStates}(Q, G)$
$\mathcal{B} \leftarrow \{q \oplus t : q \in Q,\ t \in \mathcal{T}_{\mathrm{last}(q)},\ \textsc{ValidJoin}(q, t)\}$
$\mathcal{B} \leftarrow \textsc{ConstructionRepresentatives}(\mathcal{B})$
$\mathcal{B} \leftarrow \textsc{ProtectedDestinationBudget}(\mathcal{B})$
$(\mathcal{B}, q) \leftarrow \textsc{DeduplicateAndScore}(\mathcal{B}, h_\theta)$
if $|\mathcal{B}| < 1000$ or $q_{(1000)} \leq 0$ **then**
  return unresolved ▷ retain failure provenance
end if
$\widetilde{q}(P) \leftarrow q(P)/\max_{P' \in \mathcal{B}} q(P')$
$\tau \leftarrow \min\{0.75, \widetilde{q}_{(1000)}\}$
$\mathcal{L}_s \leftarrow \{P \in \mathcal{B} : \widetilde{q}(P) \geq \tau\}$
return $\mathcal{T}_s, \mathcal{L}_s$ ▷ retain all cutoff ties and path provenance

F. Consistent Candidate Generation and Path Inference

a) Reusing the complete tree.: Training and query candidate generation both consume the same frozen $\mathcal{T}_s$. For a destination d, each anchor supplies intact forward observed tails toward d and, when available, a shortest-path proposal from the permitted topology. Every joined candidate is checked against its full prefix, endpoint, length, AS vocabulary, loop constraints, and commercial-phase continuation. The topology proposal supports candidate coverage; it is not admitted as a training label merely because it is shortest. No unrestricted generator bypasses the selected source tree.

Prefixes with fully public, commercially compatible evidence have priority over prefixes containing unknown or conflicting public roles. The latter are used for a query only if the former yield no candidate. These are ordinal evidence tiers, not route probabilities. Within the selected tier, stable deduplication and round-robin allocation across first hops and actual three-AS prefixes retain at most 200 candidates per query. Lower tiers do not fill unused higher-tier slots. This allocation preserves available forwarding alternatives without asserting a uniform real-world route distribution.

b) Learning and selecting a route.: Our implementation uses GMPI [9] as the downstream path scorer, retaining its architecture and normal-history Huber objective. The admitted $\mathcal{L}_s$ supplies surrogate training route labels and the local path knowledge consumed by candidate generation. Both consumers verify the same ordered path-set identity. Independently available observations supply historical contexts, kept separate from candidate paths at the model interface.

An internal split of the augmented supervision holds out AS pairs for checkpoint selection; paths of a pair cannot span its training and validation partitions. Final held-out source routes remain closed until evaluation. At inference, the trained model ranks $\mathcal{C}_{s,d}$ using its historical context $H_{s,d}$ and returns the highest-scoring route, or no estimate when the candidate set is empty. The implementation also handles a same-AS query with (s) and an otherwise uncovered direct public edge with (s, d); these cases remain in the fixed evaluation denominator. This separation permits augmentation and inference to be evaluated independently.

V. Evaluation

We ask three questions: how well do selected prefixes cover observed routing choices; how much complete-path support does the admitted library have; and how accurately does the final model estimate routes? The BGP setting evaluates fixed source–destination pairs. A separate domain-endpoint dataset supports a data-plane check without using those measurements as training labels.

A. Datasets and Experimental Protocol

a) Monthly BGP queries.: We use the registered February 1, March 1, and April 1, 2026 snapshots. The ten designated sources per date come from the original largest per-source path files, giving 685,820, 685,765, and 688,302 distinct query pairs, respectively. The same ten sources are registered for all three dates. Source-level results retain this cohort across snapshots. This selection favors well-observed reference sources and does not represent a random sample of Internet ASes.

For upstream supervision, the eligible source counts are 391, 404, and 409 across the three dates. They are split by source into 260/65/66, 269/67/68, and 272/68/69 training/validation/test origins, respectively; the formal ten-source cohorts are separate. Upstream folds isolate the source being predicted, and background-tail indices mask the supervised source cohorts. These boundaries differ from downstream pair-disjoint validation within one augmented source.

b) Reused-feature provenance.: The experiments reuse dated graph and AS representation artifacts, including walk embeddings and AS attributes. Their identities are recorded, but origin-level provenance for their construction is unavailable. Although final route labels are excluded from construction and checkpoint selection, the results do not establish that all reused graph-derived features are free of target-source observations. This limits their interpretation as a strictly source-withheld experiment.

c) Independent domain-endpoint measurements.: A frozen Tranco list supplies one million ranked domains [26]. Resolving it on October 15, 2026 yields 911,257 resolved domains and 388,491 unique IPv4 targets. Measurements from AS63835 include 388,491 TCP and 243,207 ICMP Paris traceroute records, collected on October 15–16. For each target, processing retains the earliest TCP observation and uses ICMP Paris results where TCP did not reach it. These records are observations, not independent AS-pair queries.

Hop addresses are mapped using a dated CAIDA prefix-to-AS table, and consecutive copies of an ASN are collapsed. An unknown run bounded by the same ASN can be removed under the same-AS gap assumption. Gaps between different ASes or at endpoints are not filled. Eligibility requires a reached target, unambiguous destination origin, matching source and destination ASes, no remaining unknown hops, and no AS loop. The resulting 22,844 eligible records yield 2,198 distinct ordered paths over 1,890 AS pairs. Endpoint domains are not individually certified HTTP services. This filter selects a cleaner, nonrepresentative subset of measured paths.

For this setting, October 15 BGP evidence supplies construction inputs, while traceroute sequences supply final reference labels only. AS63835 has no source-originated path in that BGP corpus. The PATS evaluation on these queries remains unreported.

B. Implementation and Checkpoint Selection

All reported PATS runs use a root limit of 21 in the low-ρ regime, at most ten recursive children per chosen root, a three-AS prefix limit, and a 20-AS complete-path limit. Usable-tail capacity is compared with 300; relative quality admission starts at 0.75 and requests at least 1,000 paths. Query capacity is 200. These values are part of the frozen run protocol, including runtime overrides.

The downstream model uses 256-dimensional AS representations, hidden size 512, batch size 64, Adam learning rate 0.001, and at most 30 epochs. We use normal-history GMPI with a Huber training objective. A deterministic hash split assigns 10% of augmented training pairs to validation, with all alternatives for a pair kept together. Early stopping uses three non-improving validation epochs and seed 20261015. This validation measures agreement with construction-derived labels; it is not validation against independently measured target routes. The checkpoint and input identities are frozen before the held-out reference sequences are opened for evaluation.

C. Metrics at Each Stage

a) Neighbor and prefix selection.: For pair q, let $\mathcal{Y}_q$ contain its distinct observed reference routes. First-hop coverage averages $\mathbf{1}[P_1 \in \mathcal{N}_s]$ within each pair and then across pairs, where $\mathcal{N}_s$ contains selected neighbors. Ordered prefix coverage instead checks whether a reference route begins with an enabled prefix in $\mathcal{T}_s$. This distinguishes selecting a correct first hop from preserving the right continuation after it. Paths too short to define a hop or explored prefix are excluded from the corresponding denominator. Selected

neighbors absent from a finite reference are unverified, rather than proven false.

b) Augmented-path support and branch representation.: Let $\mathcal{L}_s^Q$ contain admitted paths with a reference-covered endpoint pair. Complete-path observation support is

$$U_s = \frac{|\{P \in \mathcal{L}_s^Q : P \in \mathcal{Y}_{(s,P_{\text{last}})}\}|}{|\mathcal{L}_s^Q|}. \tag{9}$$

Support in (9) requires equality of the complete ordered sequence. Paths outside the reference-covered pairs are reported separately. We also distinguish paths already present as intact source-rooted suffixes in the permitted public corpus from newly joined paths. This prevents known-path reuse from being mistaken for evidence that an inferred join is correct.

For every selected neighbor, we report the number of admitted paths through it. Branch representation R counts selected, observed neighbors with at least one admitted path; it does not require that path to match a complete reference. A stricter diagnostic checks whether any path through the neighbor has complete ordered reference support. Coverage denominators retain selected, observed neighbors with zero output. These counts test branch representation directly: neither similar total library sizes nor a relative score gate implies balanced coverage. We do not require equal counts or interpret path frequency as traffic volume. Library support and branch representation are calculated from frozen libraries and reference sequences separately from final inference scores.

c) Final path agreement.: We preserve AS-set Jaccard as the primary metric. For AS set $S(P)$, including both endpoints, define

$$\begin{aligned} J(P,Y) &= \frac{|S(P) \cap S(Y)|}{|S(P) \cup S(Y)|}, \\ A_J &= \frac{1}{|Q|} \sum_{q \in Q} \text{round}_3\left(\max_{Y \in \mathcal{Y}_q} J(\widehat{P}_q, Y) \right). \end{aligned} \tag{10}$$

The final metric in (10) counts missing, cyclic, or endpoint-invalid predictions as zero and gives each fixed pair equal weight. Per-source Jaccard uses that source's pairs; the macro average gives each source equal weight. Ordered exact agreement additionally requires $\widehat{P}_q \in \mathcal{Y}_q$. Jaccard can be one despite a different AS order, so it is not an exact-route metric.

D. Prefix and Augmented-Library Results

Table III separates first-hop selection, branch representation, and complete-path support. In February, 34,841 of 37,884 reference-evaluable admitted paths match a complete ordered reference, giving 91.97% support. In March the corresponding count is 29,524 of 33,290, or 88.69%. There are 32 and ten admitted paths, respectively, whose AS pairs have no reference. These aggregate support rates weight each admitted path equally; equal-source means are lower, 86.39% and 83.80%, indicating variation across sources. April has 29,913 supported paths among 32,739 evaluable paths (91.37%); 22 admitted paths have no reference pair.

The libraries represent 156 of 169 selected, observed neighbors in February, 159 of 167 in March, and 158 of 168 in April. Representation is incomplete despite the minimum-library rule. The sparse-source case AS328840 further shows why first-hop coverage is insufficient: its selected first hops cover almost all observed routes, while complete-path support is only 5.93% in February, 26.00% in March, and 2.79% in April. First-hop set coverage therefore leaves both subsequent prefix choices and the paths retained through each neighbor to be assessed. These observations motivate reporting prefix and path support separately from the final query metric. They do not establish a support rate specifically for previously unobserved joins, because intact known suffixes have not been removed from this diagnostic.

E. End-to-End BGP Results

Table IV reports the full query cohorts for all three monthly snapshots. Pair-weighted Jaccard is 82.56% in February, 85.06% in March, and 83.45% in April. Equal-source averages are reported separately to expose differences in query counts. Ordered exact agreement is lower, showing that AS-set agreement and recovering an identical sequence are distinct outcomes.

Table V provides source-level detail, including selected roots and admitted path counts. Such detail matters because a single aggregate can hide a source with weak constructed supervision. For example, February AS328840 reaches only 62.78% Jaccard despite having an admitted library above the minimum. The existence of enough training paths therefore does not establish that they contain the source's relevant routing choices.

These measurements assess the complete construction-and-inference pipeline on the fixed query sets. They do not isolate the causal effect of ρ, recursion width, metadata, or quality admission. Such an ablation must hold the evidence, query identities, training split, and downstream model fixed. Comparison with other inference methods also requires the same query universe and permitted evidence.

VI. Discussion and Limitations

a) Public support is partial evidence.: Observed tails encode real advertised routing choices, but joining one to a new prefix does not prove that the source learns, selects, or forwards along the result. Non-observation is also not evidence of nonexistence. Relationship constraints reject conflicts with the adopted commercial model; they cannot explain every operational exception. Unknown relationships are preserved, which avoids invented certainty but leaves some structurally feasible candidates weakly supported.

b) Exploration and representation.: The source-shape statistic is an operational exploration rule, not a BGP theorem. A high value suggests routing diversity beyond a restricted first-hop boundary, but it does not identify the best route for each destination. A three-AS prefix cannot model choices that occur deeper in a long chain. The relative score gate can retain many similar paths, and its 1,000-path minimum may lower the effective threshold. Neither step provides a hard guarantee of representation through every important neighbor.

TABLE III

STAGE DIAGNOSTICS AGAINST DISTINCT ORDERED BGP REFERENCES. C_1 AVERAGES THE REFERENCE-PATH FIRST-HOP COVERAGE WITHIN EACH PAIR, THEN EQUALLY OVER PAIRS. R IS THE PERCENTAGE OF SELECTED, OBSERVED NEIGHBORS REPRESENTED BY AT LEAST ONE ADMITTED PATH. U IS COMPLETE ORDERED OBSERVATION SUPPORT AMONG ADMITTED PATHS WITH REFERENCE-COVERED PAIRS. AN UNOBSERVED PATH IS NOT THEREBY PROVEN UNREAL.

	2026-02-01			2026-03-01			2026-04-01		
Source AS	C_1 (%)	R (%)	U (%)	C_1 (%)	R (%)	U (%)	C_1 (%)	R (%)	U (%)
16552	86.07	95.00	99.08	87.99	100.00	96.84	87.86	95.24	97.82
19529	100.00	50.00	81.48	100.00	100.00	68.58	100.00	100.00	93.90
32590	73.73	100.00	98.97	77.43	100.00	97.63	76.05	100.00	98.51
34927	87.91	90.48	97.35	87.46	90.48	97.32	87.70	90.48	97.91
42473	66.56	94.74	97.84	58.67	88.89	95.56	58.53	94.44	97.44
49544	74.58	95.24	99.00	73.92	95.24	98.37	73.58	95.24	98.94
58057	93.77	94.74	96.24	93.72	100.00	87.46	93.50	94.74	85.67
199524	75.13	95.24	99.36	73.49	95.24	98.64	63.31	100.00	99.07
328840	99.99	50.00	5.93	99.99	100.00	26.00	99.99	66.67	2.79
398465	93.62	85.71	88.66	93.68	90.48	71.56	93.89	85.71	84.84

TABLE IV

PATHVEIL INFERENCE ON THE FIXED BGP QUERY SETS. JACCARD USES THE PER-QUERY ROUNDING IN (10). AS MEAN WEIGHTS THE TEN SOURCES EQUALLY; PAIR MEAN AND EXACT AGREEMENT WEIGHT ALL QUERIES EQUALLY.

Snapshot	Sources	Queries	Admitted paths	AS mean (%)	Pair mean (%)	Exact (%)
2026-02-01	10	685,820	37,916	82.52	82.56	52.90
2026-03-01	10	685,765	33,300	85.01	85.06	58.27
2026-04-01	10	688,302	32,761	83.41	83.45	54.76

TABLE V

PER-SOURCE NEIGHBOR SELECTION, ADMITTED PATH LIBRARIES, AND INFERENCE. N IS THE NUMBER OF SELECTED FIRST HOPS, A IS THE NUMBER OF ADMITTED AUGMENTED PATHS, AND J IS MEAN QUERY JACCARD (%).

	2026-02-01			2026-03-01			2026-04-01		
Source AS	N	A	J	N	A	J	N	A	J
16552	21	4,574	87.20	21	3,709	87.35	21	3,985	86.59
19529	3	1,420	83.75	4	1,009	82.59	4	1,000	82.86
32590	21	3,689	89.85	21	3,334	90.66	21	3,349	89.84
34927	21	4,246	81.92	21	3,514	84.23	21	3,462	84.62
42473	21	3,749	83.38	21	3,537	84.98	21	3,320	82.01
49544	21	4,286	88.51	21	4,120	86.98	21	3,663	85.69
58057	21	4,949	76.96	21	4,131	78.65	21	4,577	74.74
199524	21	4,044	88.24	21	3,240	90.59	21	3,327	89.65
328840	4	1,770	62.78	3	1,023	80.57	3	1,005	74.40
398465	21	5,189	82.64	21	5,683	83.49	21	5,073	83.65

These properties must be evaluated with the prefix and library metrics, alongside final Jaccard.

c) Learning and evaluation boundaries.: The framework depends on other networks' measurements, dated metadata, and supervised upstream sources. "Without source measurements" refers to the target source input, not to a globally measurement-free system. Current downstream validation uses augmented labels, so it can prefer a model that reproduces construction bias. The feature-provenance boundary described in the protocol limits strict cold-start interpretation. Matched baselines and component ablations are needed to establish comparative and causal benefits.

d) Scope of applications.: An inferred route can support a view of likely transit dependencies or comparison of destination networks. It cannot by itself ensure that traffic avoids an AS, certify a security policy, or predict application performance. Service deployment also depends on endpoint selection, capacity, latency, and failures. The single-source domain-endpoint dataset checks a useful deployment setting but cannot establish generalization across access networks or all Web requests.

VII. CONCLUSION

PathVeil addresses AS path inference when observations originating from the target source are unavailable as construction inputs. Prefix-anchored tail stitching combines source-

conditioned exploration, intact observed segments, and learned quality admission to create source-specific learning evidence. A shared prefix tree and admitted path library tie training to query generation. The three monthly evaluations obtain 82.56%, 85.06%, and 83.45% pair-weighted AS-set Jaccard under the documented feature-provenance boundary. The framework separates evidence support, branch representation, and final route agreement, providing explicit criteria for further validation.

REFERENCES

[1] X. Li, C. Wang, Y. Yang, C. Hou, B. Hou, and Z. Cai, "Internet interdomain path inferring: Methods, applications, and future directions," *Computers, Materials and Continua*, vol. 81, no. 1, pp. 53–78, 2024.

[2] V. Jacobson, "Traceroute." ftp://ftp.ee.lbl.gov/traceroute.tar.gz., 1988.

[3] RIPE NCC, "RIPE Atlas: User-defined measurements." https://atlas.ripe.net/docs/getting-started/user-defined-measurements/. Accessed: 2026-10-24.

[4] University of Oregon, "Route Views Project." https://www.routeviews.org/routeviews/, 2026. Accessed: 2026-07-30.

[5] RIPE NCC, "Routing Information Service (RIS)." https://www.ripe.net/analyse/internet-measurements/routing-information-service-ris/, 2026. Accessed: 2026-07-30.

[6] J. Qiu and L. Gao, "AS path inference by exploiting known AS paths," in *Proceedings of the IEEE Global Telecommunications Conference*, pp. 1–5, IEEE, 2006.

[7] Z. Jin, X. Shi, Q. Ma, L. Sun, Z. Wang, X. Yin, and J. Wu, "Which way to go? inferring fine-grained as paths with pathradar," in *IEEE INFOCOM 2025 - IEEE Conference on Computer Communications*, pp. 1–10, 2025.

[8] X. Li, Z. Cai, B. Hou, N. Liu, F. Liu, and J. Cheng, "Probinfer: Probability-based AS path inference from multigraph perspective," *Computer Networks*, vol. 180, p. 107377, 2020.

[9] X. Li, T. Zhou, Z. Cai, and J. Su, "Realizing fine-grained inference of AS path with a generative measurable process," *IEEE/ACM Transactions on Networking*, vol. 31, no. 6, pp. 3112–3127, 2023.

[10] X. Li, C. Wang, T. Yang, A. Shen, Z. Qiu, B. Hou, and Z. Cai, "Realizing personalized and adaptive inference of AS paths with a generative and measurable process," *IEEE Transactions on Networking*, vol. 33, no. 2, pp. 729–744, 2025.

[11] Y. Rekhter, T. Li, and S. Hares, "A border gateway protocol 4 (BGP-4)," RFC 4271, RFC Editor, Jan. 2006.

[12] CAIDA, "AS Relationships." https://www.caida.org/catalog/datasets/as-relationships/. Accessed: 2026-10-24.

[13] CAIDA, "AS Rank." http://as-rank.caida.org/, 2023.

[14] PeeringDB, "Peeringdb: The interconnection database." https://www.peeringdb.com/, 2026. Accessed: 2026-07-19.

[15] L. Gao, "On inferring autonomous system relationships in the internet," *IEEE/ACM Trans. Netw.*, vol. 9, no. 6, pp. 733–745, 2001.

[16] B. Augustin, X. Cuvellier, B. Orgogozo, F. Viger, T. Friedman, M. Latapy, C. Magnien, and R. Teixeira, "Avoiding traceroute anomalies with paris traceroute," in *Proceedings of Internet Measurement Conference*, (New York, NY, USA), pp. 153–158, ACM, 2006.

[17] M. Luckie, "Scamper: A scalable and extensible packet prober for active measurement of the internet," in *Proceedings of the 10th ACM SIGCOMM Conference on Internet Measurement*, pp. 239–245, ACM, 2010.

[18] RIPE NCC, "RIPE Atlas: Measurement result format, version 5000." https://atlas.ripe.net/docs/apis/measurement-result-format/version-5000/. Accessed: 2026-10-24.

[19] Z. M. Mao, J. Rexford, J. Wang, and R. H. Katz, "Towards an accurate AS-level traceroute tool," in *Proceedings of the ACM SIGCOMM 2003 Conference*, pp. 365–378, ACM, 2003.

[20] H. V. Madhyastha, T. Isdal, M. Piatek, C. Dixon, T. Anderson, A. Krishnamurthy, and A. Venkataramani, "iPlane: An information plane for distributed services," in *Proceedings of the USENIX Symposium on Operating Systems Design and Implementation*, pp. 367–380, USENIX, 2006.

[21] H. V. Madhyastha, E. Katz-Bassett, T. E. Anderson, A. Krishnamurthy, and A. Venkataramani, "iPlane nano: Path prediction for peer-to-peer applications.," in *Proceedings of the USENIX Symposium on Networked Systems Design and Implementation*, vol. 9, pp. 137–152, USENIX, 2009.

[22] R. Singh, D. Tench, P. Gill, and A. McGregor, "PredictRoute: A network path prediction toolkit," *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 5, no. 2, 2021.

[23] T. Wu, J. H. Wang, J. Wang, and S. Zhuang, "Routeinfer: Inferring interdomain paths by capturing isp routing behavior diversity and generality," in *Proceedings of the International Conference on Passive and Active Measurement*, (Berlin, Heidelberg), pp. 216–244, Springer, 2022.

[24] K. Schomp, O. Bhardwaj, E. Kurdoglu, M. Muhaimen, and R. K. Sitaraman, "Akamai DNS: Providing authoritative answers to the world's queries," in *Proceedings of the ACM SIGCOMM 2020 Conference*, (New York, NY, USA), pp. 465–478, ACM, 2020.

[25] T. Koch, K. Li, C. Ardi, E. Katz-Bassett, M. Calder, and J. Heidemann, "Anycast in context: A tale of two systems," in *Proceedings of the ACM SIGCOMM 2021 Conference*, (New York, NY, USA), pp. 398–417, ACM, 2021.

[26] V. Le Pochat, T. Van Goethem, S. Tajalizadehkhoob, M. Korczyński, and W. Joosen, "Tranco: A research-oriented top sites ranking hardened against manipulation," in *Proceedings of the Network and Distributed System Security Symposium*, 2019.

[27] CAIDA, "Routeviews Prefix to AS mappings Dataset for IPv4 and IPv6." https://www.caida.org/catalog/datasets/routeviews-prefix2as/. Accessed: 2026-10-24.

[28] "Center for Applied Internet Data Analysis." https://www.caida.org/, 2023.

[29] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, pp. 3146–3154, Curran Associates, Inc., 2017.